ACADEMIC PROJECT PRESENTATION

Problem Statement: PS-1

AI-Based Personalized Learning Content Generator

Adaptive educational engine dynamically adjusting notes, explanations, and assessments.

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ENROLLMENT & CLASS

3rd Semester AIML

CONTEXT & CHALLENGE

Problem Statement

Static Educational Content

Traditional materials cannot adapt to individual student speed.

Persistent Knowledge Gaps

Students require customized explanations and targeted practice questions.

Proposed AI Intervention

An automated system adjusting difficulty based on proficiency.

ARCHITECTURE OVERVIEW

Proposed Solution

Flask-Based Web Engine

Lightweight application generating personalized learning assets instantly.

Structured User Inputs

Topic, Subject, Baseline Proficiency Level, and Quiz Score.

Dynamic Curriculum Calibration

Difficulty levels update automatically from performance metrics.

GENERATED DELIVERABLES

- **Tailored Conceptual Explanations**
- **Structured Revision Notes**
- **Targeted Practice Questions**

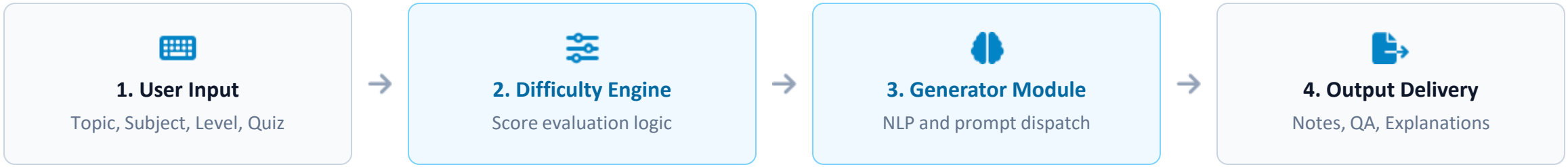
SCOPE & INTENT

Project Objectives

- **Instant Generation:** Produce customized learning content in real time.
- **Adaptive Calibration:** Adjust difficulty based on student comprehension.
- **Complete Toolkit:** Provide explanations, notes, and practice queries.
- **Applied NLP:** Implement methods suitable for 3rd-semester AIML level.
- **Efficient Architecture:** Leverage rule-based logic and external GenAI APIs.

PIPELINE FLOW

System Architecture



Unified Presentation Layer: Web Interface implemented using Flask, HTML5, and CSS3

DEVELOPMENT STACK

Technologies Used

Python

Core programming logic.

Flask

Backend web framework.

HTML & CSS

Frontend user interface.

Rule-based NLP

Template extraction logic.

GenAI APIs

Optional cloud expansion.

ADAPTIVE INTELLIGENCE

Difficulty Adjustment Mechanism

Score: 80 - 100%

Level +1

Advance difficulty to the next tier.

Score: 40 - 79%

Maintain

Reinforce current mastery tier.

Score: 0 - 39%

Level -1

Ease difficulty for foundational review.

Progression Path: Tier 1: Beginner › Tier 2: Intermediate › Tier 3: Advanced

TEMPLATE ENGINEERING

Prompt Design

Tier 1

Beginner

Simple definitions, direct examples, and fundamental questions.

Tier 2

Intermediate

Workflow processes, advantages, constraints, and application queries.

Tier 3

Advanced

Optimization patterns, complexity analysis, edge cases, and diagnostics.

GENERATION SHOWCASE

Sample Output - Beginner

Parameters

Python Loops

Assigned Level: Beginner

Generated Output Format

Simple explanation of loops, fundamental notes, and easy questions.

Sample Assessment Question

"Define Python loops in your own words."

GENERATION SHOWCASE

Sample Output - Intermediate

Parameters

Machine Learning

Assigned Level: Intermediate

Generated Output Format

Working process explanation, system benefits, constraints, and applied tasks.

Sample Assessment Question

"Explain the working process of machine learning with an example."

GENERATION SHOWCASE

Sample Output - Advanced

Parameters

Data Structures

Assigned Level: Advanced

Generated Output Format

Optimization-focused concepts, complexity notes, and higher-order analysis.

Sample Assessment Question

"Analyze the efficiency of data structures for a large dataset."

SYSTEM BENEFITS

Advantages

Personalized Learning

Tailors instruction directly to student understanding.

Instant Generation

Zero waiting latency for study materials.

Simple Interface

Clean, responsive, and student-friendly interaction.

Progressive Growth

Enables smooth transition through difficulty tiers.

Extensible Stack

Ready for plug-and-play generative AI API hookups.

ROADMAP

Future Scope

- **API Integration:** Direct integration with OpenAI or Gemini API models.
- **User Management:** Student login portal and historical progress tracking.
- **Automated Scoring:** Real-time quiz evaluation and performance analysis.
- **Multilingual Delivery:** Dynamic translation and localized content generation.
- **Resource Curation:** Personalized external reading and video recommendations.

SUMMARY

Conclusion

Core Demonstration

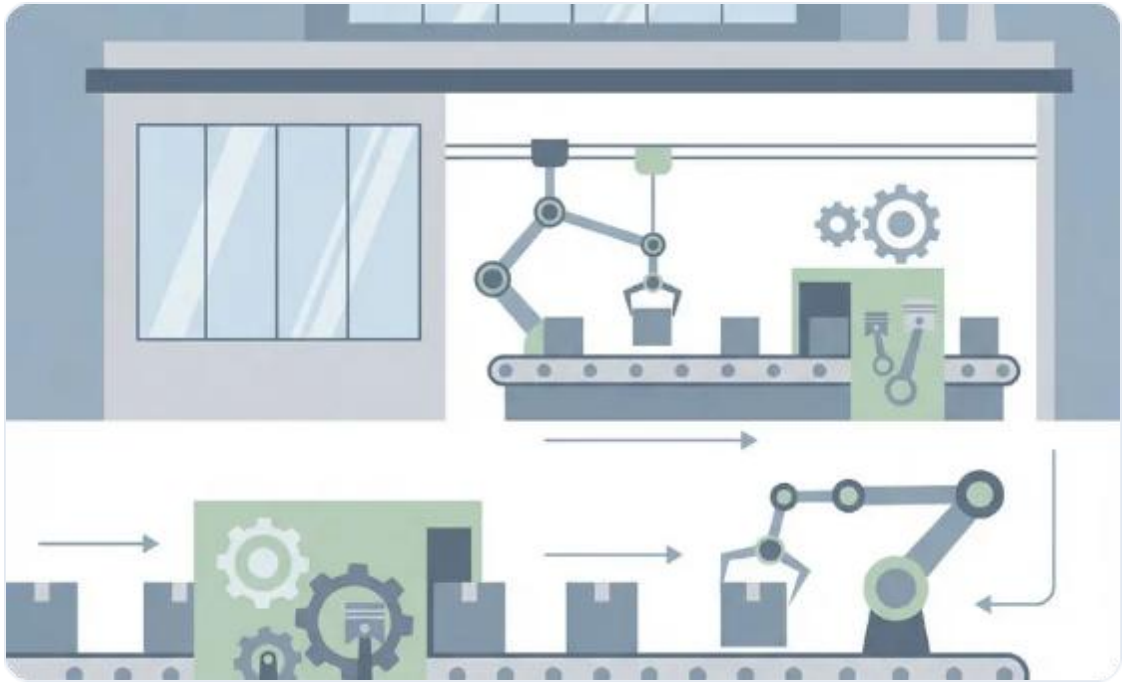
Successfully demonstrates dynamic AI-assisted learning content delivery.

Dynamic Calibration

Accurately updates content difficulty via continuous performance feedback.

Production Readiness

Practical, scalable, and fully prepared for active classroom adoption.



END OF PRESENTATION

Thank You

3rd Semester AIML